Guarani Mbya Writing Assistant

Short project description for a low-resource language fine-tuning project

1. What is this project about?

This project is about creating a small AI writing assistant for a language with limited digital data. The inspiration is Guarani Mbya, an Indigenous language from Brazil. The assistant should help a user write by suggesting words or continuing a sentence. The project is not only technical: it also asks how to use language data in a respectful and responsible way.

2. Project goal

The goal is to test whether a small language model can be fine-tuned with only a small dataset and still give useful writing suggestions.

- **Technical goal:** build a working fine-tuned model and a simple demo.
- **Research goal:** check if the model actually learns language patterns or only memorizes the training data.
- **Ethical goal:** explain how language ownership, consent, and community control should be handled.

3. Why this project matters

Many smaller languages are spoken in daily life, but they are not well supported by digital tools such as keyboards, autocorrect, and AI assistants. When people write online, they may switch to a dominant language because it is easier. This project explores whether AI can support writing in smaller languages without taking control away from the people who speak them.

4. Project scope

- Use Guarani Mbya as the real-world inspiration, but do not use sensitive Indigenous language data unless it is clearly safe and allowed.
- Work with a small dataset of around 1,000 to 5,000 examples.
- Use a small pre-trained model, for example GPT-2 small, BLOOM-560m, or another lightweight open model.
- Use efficient fine-tuning, such as LoRA or QLoRA, so the training can work with limited hardware.
- Build a simple text-based demo where a user types something and the model gives a suggestion.
- Evaluate the model with examples that were not used during training.

5. What should be built?

- A cleaned small dataset with a short explanation of its source.
- A fine-tuning script or notebook that shows how the model was trained.
- A small demo interface, for example with Streamlit.
- An evaluation section that compares useful output, bad output, and memorized output.
- A short report or decision log explaining choices, results, risks, and limitations.

6. How to start

Choose the language setup: Use a safe public low-resource dataset, or simulate the low-resource situation by limiting the data of another language.

Collect and clean data: Prepare 1,000 to 5,000 examples. Remove duplicates, broken text, and examples that should not be used.

Select a small model: Choose a model that can realistically run on student hardware.

Fine-tune the model: Train the model on the small dataset. Use LoRA or QLoRA if full fine-tuning is too heavy.

Build the demo: Make a simple screen with a text box and an output area for suggestions.

Evaluate carefully: Test with new examples. Check whether the model creates useful new text or repeats training examples.

Write the final explanation: Explain what worked, what failed, and what would be needed before real community use.

7. Simple explanation of key terms

- **Low-resource language:** a language with little digital text for AI training.
- **Fine-tuning:** training an existing AI model more for one specific task or language.
- **LoRA / QLoRA:** lighter ways to fine-tune a model without needing very powerful hardware.
- **Memorization:** the model copies training examples instead of learning patterns.
- **Generalization:** the model can handle new examples it did not see before.
- **Community consent:** the people connected to the language understand the project and can agree, refuse, or pause it.

8. Evaluation plan

The evaluation should answer one main question: did the model learn something useful, or did it only copy the dataset?

- Keep some examples outside the training data and use them only for testing.
- Compare model outputs with the training data to detect copying.
- Show a few good examples and a few bad examples.
- Explain the limitations honestly, especially if no native speaker or domain expert could review the output.

9. Ethical focus

The project should not claim to “save” a language. It should be presented as a prototype. Real use with an Indigenous community would require trust, permission, and shared control. The most important lesson from the Guarani Mbya case is that the community must have the power to say “wait” or “stop”.

10. Final expected result

At the end, the project should deliver a small but complete AI workflow: dataset preparation, fine-tuning, demo, evaluation, and ethical reflection. A foreign person reading the report should understand what was built, why it was built, how it was tested, and what the responsible limits are.